cascading plants, circuits, stability

1. Cascading Plants
   - signals and ordinary differential equations
   - bandpass as lowpass followed by highpass
2. Electric Circuits
   - surge impedance of a device
   - lowpass and highpass filters
3. Stability
   - the poles of the transfer function
   - two vehicle traffic

MCS 472 Lecture 33
Industrial Math & Computation
Jan Verschelde, 6 April 2026

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plants and convolutions

Definition (plant)

A *plant* is a linear, time invariant, causal map, which maps signals into signals.

Examples: filters, mechanisms, circuits.

By the linearity, time invariance, and causality, the output $y = h \star u$ where h is the response to the unit impulse and u is the input.

Definition (convolutions of functions)

Let f and g be functions of t. The *convolution* $f \star g$ of f and g is

$$f \star g = \int_0^t f(\tau) g(t - \tau) d\tau.$$

operational calculus

The differential equation

$$\frac{d}{dt}y(t) + y(t) = u(t), \quad y(0) = 0$$

is transformed into

$$sY(s) + Y(s) = U(s) \quad \Rightarrow \quad Y(s) = \left(\frac{1}{s+1} \right) U(s),$$

where $Y(s) = \mathcal{L}\{y\}$ and $U(s) = \mathcal{L}\{u\}$.

In the time domain, the solution is then

$$y(t) = e^{-t} \star u(t) = \int_0^t e^{-\tau} u(t-\tau) d\tau.$$

a computational experiment

Take two signals, one at low, another at high frequency.

- $u_{\rm low}(t) = \sin(2\pi t/1000)$

  $u_{\rm low}$ takes 1000 seconds to complete one cycle.

- $u_{\rm high}(t) = \sin(2\pi t 1000)$

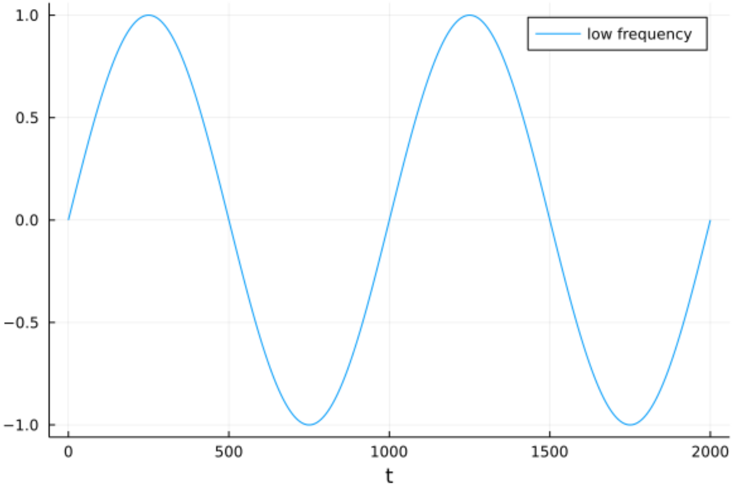
  $u_{\rm high}$ does 1000 cycles in one second.

Consider then the solution $y(t)$ of

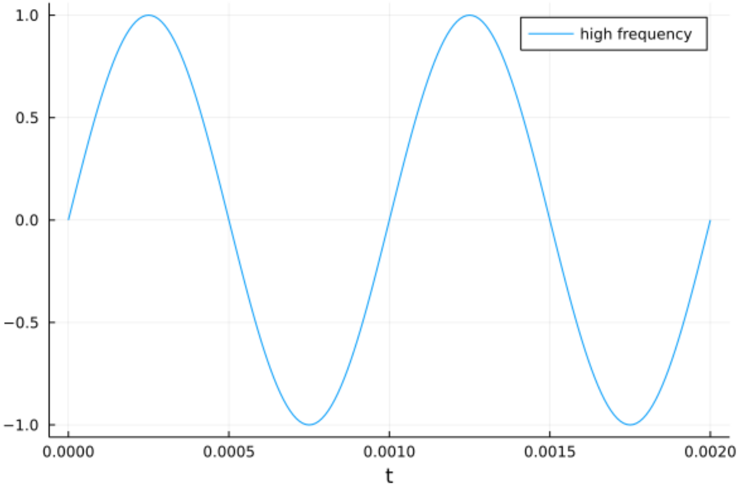
$$\frac{d}{dt} y(t) + y(t) = u(t), \quad y(0) = 0,$$

for $u = u_{\rm low}$ and $u = u_{\rm high}$.

a low frequency signal $\sin(2\pi t/1000)$



a high frequency signal $\sin(2\pi t 1000)$



using `SymPy.jl` in a Jupyter notebook

We have the low frequency and the high frequency signals:

```
lowfreq(t) = sin(2*pi*t/1000)
highfreq(t) = sin(2*pi*t*1000)
```

Now define the left of the ODE and the initial condition:

```
t = Sym("t")
y = SymFunction("y")
leftode = diff(y(t), t) + y(t)
initcond = Dict(y(0) => 0)
```

Then the right hand side for the low frequency signal:

```
filtlow = Eq(leftode, lowfreq(t))
outlow = dsolve(filtlow, ics=initcond)
```

output of the filter for low and high frequencies

The output for $u(t) = u_{\mathrm{low}}(t) = \sin(2\pi t/1000)$:

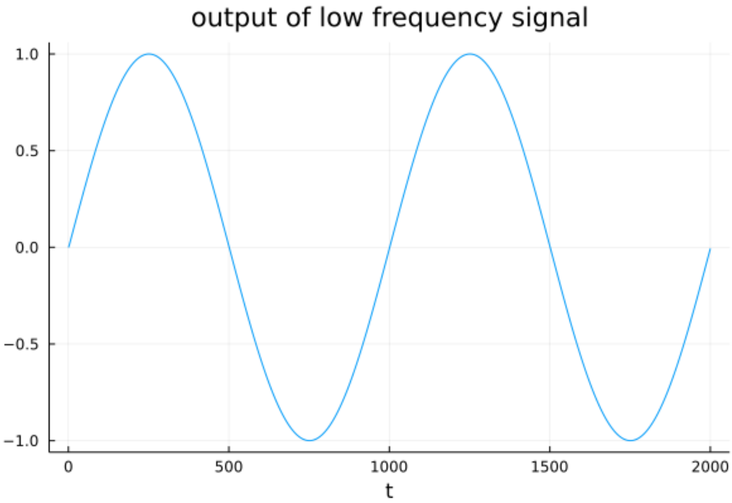
$$\begin{aligned} y(t) &= 0.99996052314088 \sin(0.00628318530717959t) \\ &\quad - 0.00628293726675839 \cos(0.00628318530717959t) \\ &\quad + 0.00628293726675839 e^{-t} \end{aligned}$$

```
filthigh = Eq(leftode, highfreq(t))
outhigh = dsolve(filthigh, ics=initcond)
```

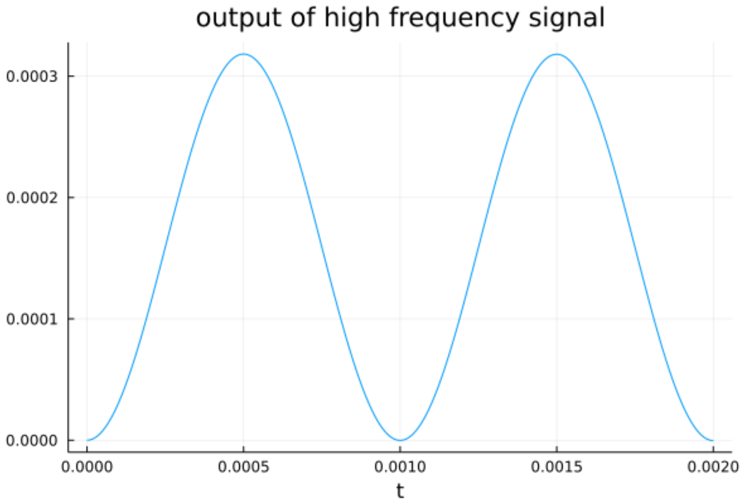
The output for $u(t) = u_{\mathrm{high}}(t) = \sin(2\pi t 1000)$:

$$\begin{aligned} y(t) &= 2.53302952689605\mathrm{E}{-8} \sin(6283.18530717959t) \\ &\quad - 0.000159154939060454 \cos(6283.18530717959t) \\ &\quad + 0.000159154939060454 e^{-t} \end{aligned}$$

output of the low frequency signal $\sin(2\pi t/1000)$



output of the high frequency signal $\sin(2\pi t 1000)$



a highpass filter

The differential equation

$$\frac{d}{dt}y(t) + y(t) = \frac{d}{dt}u(t), \quad y(0) = 0$$

is transformed into

$$sY(s) + Y(s) = sU(s) \quad \Rightarrow \quad Y(s) = \left(\frac{s}{s+1} \right) U(s),$$

where $Y(s) = \mathcal{L}\{y\}$ and $U(s) = \mathcal{L}\{u\}$.

Exercise 1:

- Solve the above ODE for $u = u_{\rm low}$ and $u = u_{\rm high}$.
- Verify that the ODE represents a highpass filter.

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a bandpass filter

Consider $u(t)$ as input to

$$\frac{d}{dt}y(t) + y(t) = u(t), \quad y(0) = 0,$$

and then give $y(t)$ as input to

$$\frac{d}{dt}z(t) + z(t) = \frac{d}{dt}y(t), \quad z(0) = 0,$$

which gives a bandpass filter.

The frequency domain representation:

$$Z(s) = \left(\frac{s}{s+1}\right)\left(\frac{1}{s+1}\right)U(s), \text{ as } Y(s) = \left(\frac{1}{s+1}\right)U(s),$$

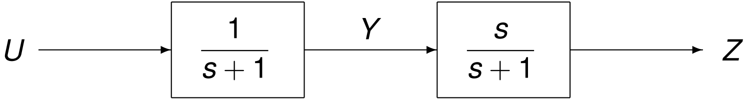
with $Y(s) = \mathcal{L}\{y\}$, $U(s) = \mathcal{L}\{u\}$, and $Z(s) = \mathcal{L}\{z\}$.

lowpass followed by highpass

The frequency domain representation

$$Z(s) = \left(\frac{s}{s+1} \right) \left(\frac{1}{s+1} \right) U(s),$$

of the bandpass filter



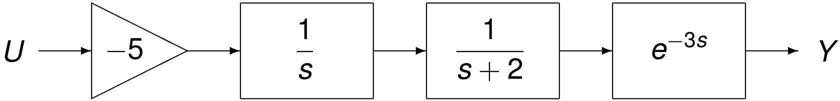
is an example of two *cascading plants*.

The transfer function of plants in cascade is the product of the transfer functions of the components in the cascade.

cascading plants

Consider a plant composed of

- an inverting linear amplifier of gain -5,
- followed by an integrator $\int_0^t$,
- followed by a convolution with e^{-2t},
- followed by a 3-second delay.



The transfer function is: $\displaystyle \frac{Y(s)}{U(s)} = -5 \left(\frac{1}{s} \right) \left(\frac{1}{s+2} \right) e^{-3s}$.

Exercise 2: Compute the time domain version of this cascading plant.

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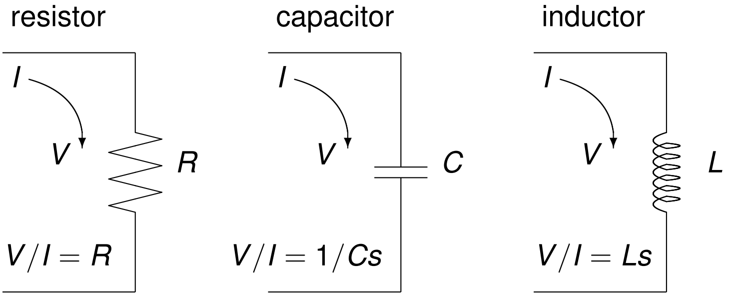
relation of voltage to current

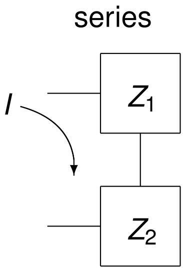
The relation between the voltage $v(t)$ across, and the current $i(t)$ through a resistor, capacitor, and inductor is shown below.



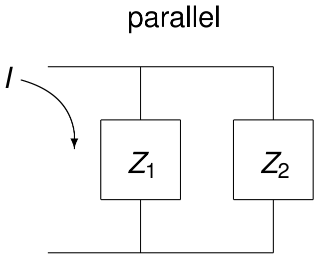
Definition (surge impedance of a device)

The ratio $Z = V/I$ is the *surge impedance* of a device.

series and parallel circuits



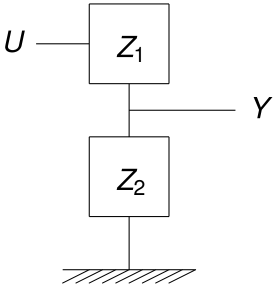
$$Z = V/I = Z_1 + Z_2$$



$$Z = V/I = \frac{1}{1/Z_1 + 1/Z_2}$$

the transfer function of a network

The network



has transfer function

$$\frac{Y}{U} = \frac{Z_2}{Z_1 + Z_2}.$$

Exercise 3: Prove that this network has this transfer function.

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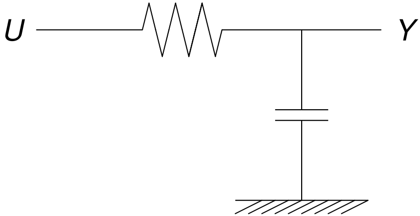
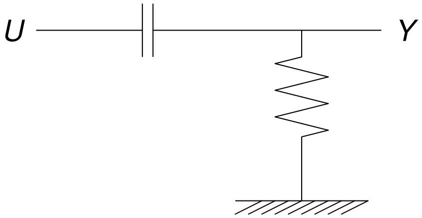
a realization of a lowpass filter



For the resistor $Z_1 = R$, and the capacitor $Z_2 = \frac{1}{Cs}$, the transfer function is

$$\frac{Y}{U} = \frac{\frac{1}{Cs}}{R + \frac{1}{Cs}} = \frac{1}{RCs + 1}$$

a realization of a highpass filter



For the capacitor $Z_1 = \frac{1}{Cs}$, and the resistor $Z_2 = R$, the transfer function is

$$\frac{Y}{U} = \frac{R}{R + \frac{1}{Cs}} = \frac{RCs}{RCs + 1}$$

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stability and poles

Definition (stability of a plant)

A plant is *stable* if bounded inputs give bounded outputs.

The inpulse response of a stable plant is decaying.

Definition (poles of a rational function)

Consider the quotient $q = f(x)/g(x)$ of two polynomials f and g.
The *poles* of q are the roots of the denominator $g(x) = 0$.

a criterion on the stability

Theorem (stability criterion)

Let the plant P have transfer function $P(s) = f(s)/g(s)$.

P is stable if and only if all poles lie in the open left complex plane:

$$\text{for any pole } z,\ g(z) = 0\text{: } Re(z) < 0.$$

Example: consider

$$P(s) = \frac{3s^2 + 2s + 1}{s^2 + 7s + 4}.$$

The poles are $-7/2 \pm \sqrt{33}/2$, so P is stable.

Exercise 4: Let $g(s) = as^2 + bs + c$, with $a, b, c > 0$.
What is the sign of the real parts of the roots of g?

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two vehicle traffic

Let $u(t)$ be the position of the leading vehicle,
and $y(t)$ be the position of the trailing vehicle.

Modeling with an ordinary differential equation, with delay τ and α:

$$\frac{d^2}{d t^2} y(t + \tau) = \alpha \left(\frac{d}{d t} u(t) - \frac{d}{d t} y(t) \right),$$

subject to the initial conditions $y(0) = y'(0) = u(0) = u'(0) = 0$.

Integration leads to the differential equation:

$$\frac{d}{d t} y(t + \tau) + \alpha y(t) = \alpha u(t).$$

In the frequency domain: $Y(s) = \dfrac{\alpha e^{-\tau s}}{s + \alpha e^{-\tau s}} U(s)$.

stability of following cars

Proposal for a Topic of a Project:

Consider the model for two vehicle traffic

$$\frac{d}{dt}y(t+\tau) + \alpha y(t) = \alpha u(t).$$

- Obtain the stability condition for this problem.

  In particular, what is the relation between τ and α for this problem to be stable?

- Show that even when stability is satisfied, a long platoon of vehicles is inherently unsafe because changes in speed of the lead car produce larger and larger deviations for cars farther down the platoon.

summary and bibliography

We illustrated the relationship between filters and ordinary differential equations and considered realizations with circuits.
The stability of a plant depends on the poles of the transfer function.

We covered sections 3, 4, and 5 of Chapter 10 of our text book.

- Charles R. MacCluer:
  Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.

An introductory text book in control has many examples to engineering:

- Richard C. Dorf and Robert H. Bishop:
  Modern Control Systems. Nineth Edition, Prentice Hall, 2001.